

Sophia Tang

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EDUCATION

University of Pennsylvania

Dual Degree with Bachelor of Science in Engineering and Bachelor of Science in Economics

Jerome Fisher Program in Management & Technology; Major in Computer Science & Statistics

Philadelphia, PA

Sept 2023 - May 2027 (*expected*)

RESEARCH EXPERIENCE

Kempner Institute, Harvard University.

Visiting Researcher. **Advised by Dr. Michael Albergo.**

- Developing theoretical generative modeling frameworks.

Boston, MA

May 2026 - Present

Chatterjee Lab, University of Pennsylvania.

Undergraduate Researcher. **Advised by Dr. Pranam Chatterjee.**

- Developing theoretical ML frameworks for generative modeling with applications in biological design and simulation.
- **First-author papers published at ICML 2025, ICLR 2026, and NeurIPS 2026 (see below).**

Philadelphia, PA

Sept 2024 - Present

Mitchell Lab, University of Pennsylvania.

Undergraduate Researcher. **Advised by Dr. Michael Mitchell.**

- Developed novel targeted lipid nanoparticle (LNP)-encapsulated mRNA therapeutics with high transport across the blood-brain barrier after systemic administration and transfection into brain cell types.
- **Several published co-authored papers and a first-author review paper (see below).**

Philadelphia, PA

Sept 2023 - Present

MACHINE LEARNING PUBLICATIONS (*denotes equal contribution; † denotes advising role)

[1] **PepTune: De Novo Generation of Therapeutic Peptides with Multi-Objective-Guided Discrete Diffusion.**

Sophia Tang*, Yinuo Zhang*, Pranam Chatterjee†.

International Conference on Machine Learning (ICML 2025), Dec 2024. <https://openreview.net/forum?id=FQoy1Y1Hd8>

[2] **Branched Schrödinger Bridge Matching.**

Sophia Tang, Yinuo Zhang, Alexander Tong†, Pranam Chatterjee†.

International Conference on Learning Representations (ICLR 2026). Jun 2025. <https://doi.org/10.48550/arXiv.2506.09007>

[3] **Foundations of Schrödinger Bridges for Generative Modeling**

Sophia Tang

Preprint. Mar 2026. <https://doi.org/10.48550/arXiv.2603.18992>

[4] **Discrete Beckmann Transport Models for One-Step Language Modeling and Reasoning.**

Sophia Tang and Shiyi Wang.

Preprint. Sept 2026. <https://doi.org/10.48550/arXiv.2609.15903>

[5] **Expanding Flow Maps**

Sophia Tang and Pranam Chatterjee†.

Advances in Neural Information Processing Systems (NeurIPS 2026). July 2026. <https://doi.org/10.48550/arXiv.2607.21585>

[6] **TR2-D2: Tree Search Guided Trajectory-Aware Fine-Tuning for Discrete Diffusion.**

Sophia Tang*, Yuchen Zhu*, Molei Tao†, Pranam Chatterjee†.

Preprint. Sept 2025. <https://doi.org/10.48550/arXiv.2509.25171>

[7] **Entangled Schrödinger Bridge Matching.**

Sophia Tang, Yinuo Zhang, Pranam Chatterjee†.

Advances in Neural Information Processing Systems (NeurIPS 2026). Oct 2025. <https://doi.org/10.48550/arXiv.2511.07406>

[8] **A2D2: Finetuning Any-Length Discrete Diffusion for Adaptive Decoding.**

Sophia Tang, Yuchen Zhu, Molei Tao†, Pranam Chatterjee†.

Preprint. June 2026. <https://doi.org/10.48550/arXiv.2606.13565>

[9] **Gumbel-Softmax Flow Matching with Straight-Through Guidance for Controllable Biological Sequence Generation.**

Sophia Tang, Yinuo Zhang, Alexander Tong†, Pranam Chatterjee†.

Preprint. Mar 2025. <https://doi.org/10.48550/arXiv.2503.17361>

[10] **mRNAutilus: Multi-Objective-Guided Discrete Generation of mRNA with Optimized Therapeutic Properties.**

Sawan Patel*, **Sophia Tang***, Yesol Kim*, Yinuo Zhang, Divya Srijay, Ping-Jung Lin, Shambhavi Shubham, Fengmei Pi, Cedric Wu, Sherwood Yao†, and Pranam Chatterjee†.

Preprint. Jun 2026. <https://doi.org/10.48550/arXiv.2605.31296>

[11] PeptiVerse: A Unified Platform for Therapeutic Peptide Property Prediction.

Yinuo Zhang, [Sophia Tang](#), Tong Chen, Elizabeth Mahood, Sophia Vincoff and Pranam Chatterjee†.

Nature Communications. Jan 2026. <https://www.biorxiv.org/content/10.64898/2025.12.31.697180v1>

[12] TD3B: Transition-Directed Discrete Diffusion for Allosteric Binder Generation.

Hanqun Cao*, Aastha Pal*, [Sophia Tang](#), Yinuo Zhang, Jingjie Zhang, Pheng Ann Heng, Pranam Chatterjee†.

International Conference on Machine Learning (ICML 2026). May 2026. <https://arxiv.org/abs/2605.09810>

[13] Multi-Objective-Guided Discrete Flow Matching for Controllable Biological Sequence Design.

Tong Chen, Yinuo Zhang, [Sophia Tang](#), and Pranam Chatterjee†.

Preprint. May 2025. <https://doi.org/10.48550/arXiv.2505.07086>

[14] Active Flow Expansion for Out-of-Distribution Discovery: from Theory to Molecules.

Riccardo De Santi, Bruce Lee, Cristian Perez Jensen, Kimon Protopoulos, [Sophia Tang](#), Cheng-Hao Liu, Pranam Chatterjee, Yisong Yue, Andreas Krause.

Advances in Neural Information Processing Systems (NeurIPS 2026). June 2026. <https://doi.org/10.48550/arXiv.2606.08802>

[15] Beckmann Transport Models: From Autonomous Flows to One-Step Maps.

Lee Cheuk-Kit, Florentin Coeurdoux, Yuyuan Chen, [Sophia Tang](#), Peter Potapchik, Yilun Du†, Michael Samuel Albergot†, Eric Vanden-Eijnden†.

Advances in Neural Information Processing Systems (NeurIPS 2026). July 2026. <https://doi.org/10.48550/arXiv.2608.01692>

BIOENGINEERING PUBLICATIONS († denotes advising role)

[16] Peptide-functionalized nanoparticles for brain-targeted therapeutics.

[Sophia Tang](#), Emily L. Han, and Michael J. Mitchell†.

Drug Delivery and Translational Research, Springer Nature. Mar 2025. <https://doi.org/10.1007/s13346-025-01840-w>

[17] Peptide-Functionalized Lipid Nanoparticles for Targeted Systemic mRNA Delivery to the Brain.

Emily L. Han, [Sophia Tang](#), Dongyoon Kim, Amanda M. Murray, Kelsey L. Swingle, Alex G. Hamilton, Kaitlin Mrksich, Marshall S. Padilla, Rohan Palanki, Jacqueline J. Li, and Michael J. Mitchell†.

ACS Nano Letters. Dec 2024. <https://doi.org/10.1021/acs.nanolett.4c05186>

[18] Predictive High-Throughput Platform for Dual Screening of mRNA Lipid Nanoparticle Blood–Brain Barrier Transfection and Crossing.

Emily L. Han, Marshall S. Padilla, Rohan Palanki, Dongyoon Kim, Kaitlin Mrksich, Jacqueline J. Li, [Sophia Tang](#), Il-Chul Yoon, and Michael J. Mitchell†.

ACS Nano Letters. Jan 2024. <https://doi.org/10.1021/acs.nanolett.3c03509>

[19] Optimized microfluidic formulation and organic excipients for improved lipid nanoparticle-mediated genome editing.

Rohan Palanki, Emily L. Han, Amanda M. Murray, Robin Maganti, [Sophia Tang](#), Kelsey L. Swingle, Dongyoon Kim, Hannah Yamagata, Hannah C. Safford, Kaitlin Mrksich, William H. Peranteau, and Michael J. Mitchell†.

Lab on a Chip, Royal Society of Chemistry. Jan 2024. <https://doi.org/10.1039/D4LC00283K>

[20] High-Throughput In Vivo Screening Identifies Structural Factors Driving mRNA Lipid Nanoparticle Delivery to the Brain.

Emily L. Han, Dongyoon Kim, Amanda M. Murray, Kaitlin Mrksich, Alex G. Hamilton, [Sophia Tang](#), Soyeon Yoo, Audrey T. Zhu, Elaine R. Tong, Rohan Palanki, Michelle L. Hall, Sean K. Bedingfield, Michael J. Mitchell†.

ACS Nano. Jan 2026. <https://doi.org/10.1021/acsnano.5c19138>

TECHNICAL ARTICLES

A Complete Guide to Spherical Equivariant Graph Transformers. <https://doi.org/10.48550/arXiv.2512.13927>

A 99-page breakdown of the theory behind spherical equivariant (SE(3)) graph neural networks (EGNNs).

A Complete Guide to Protein Folding Prediction. <https://alchemybio.substack.com/p/a-complete-guide-to-protein-folding>

A 3-hour breakdown of the three-track RoseTTAFold protein prediction model that leverages multi-modal deep learning architectures.

INVITED TALKS

Discrete Diffusion Reading Group. Sept 2026. <https://youtu.be/hfM0ULRC3qA?si=SmTKfhh2tX0JhxeX>

Microsoft Research NE Generative Modeling Seminar. July 2026. <https://youtu.be/KbZ2ekmZO90?si=SNncY1kFfyXnmizW>

Discrete Diffusion Reading Group. Nov 2025. https://youtu.be/NX2Mth_Epvk?si=LsVS_bhzyD-RV03m

Starkly Speaking Reading Group. Jul 2025. <https://youtu.be/inVYA0pQ4Wg?si=mdwhTE3wu0vXJ4ZX>

Learning on Graphs and Geometry Reading Group. Jan 2025. <https://youtu.be/KVr8ryclwdA?si=RdRPdqNqymbk3Og>

AWARDS AND HONOURS

CRA Outstanding Undergraduate Researcher Award. Jan 2026. 1 of 8 awardees across North America. The award is widely regarded as the highest honor for undergraduate computer science students and recognizes exceptional research accomplishments in computing.

<https://www.seas.upenn.edu/stories/sophia-tang-wins-cra-outstanding-undergraduate-research-award/>